# Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning

NVIDIA

**TAKEAWAY:** Nemotron 3 Super achieves up to 2.2× and 7.5× higher inference throughput than GPT-OSS-120B and Qwen3.5-122B, respectively, while achieving comparable accuracy on common benchmarks.

## 1. PROBLEM & MOTIVATION

- Large language models (LLMs) face trade-offs between accuracy, inference cost, and latency.
- Existing Mixture-of-Experts (MoE) architectures often neglect hardware constraints like memory bandwidth and communication overhead during inference.

### OUR APPROACH

- Uses a hybrid Mamba-Attention architecture with sparse LatentMoE scaling.
- Incorporates Multi-Token Prediction (MTP) layers for speculative decoding and inference acceleration.
- Pre-trains in NVFP4 precision with a 2-phase training process on 25 trillion tokens.
- Employs a periodic hybrid interleaving pattern of Mamba and attention layers.

### KEY CAPABILITIES

- Achieves up to 2.2× higher inference throughput than GPT-OSS-120B.
- Achieves up to 7.5× higher inference throughput than Qwen3.5-122B.
- Supports up to 1 million context length.

### MODEL OVERVIEW

| | | |
|---|---|---|
| **120B** Total Parameters | **12B** Active Parameters | **25T** Pre-training Tokens |
| **2.2×** Throughput vs. GPT-OSS-120B | **7.5×** Throughput vs. Qwen3.5-122B | **1M** Max Context Length |

## 2. METHOD / ARCHITECTURE

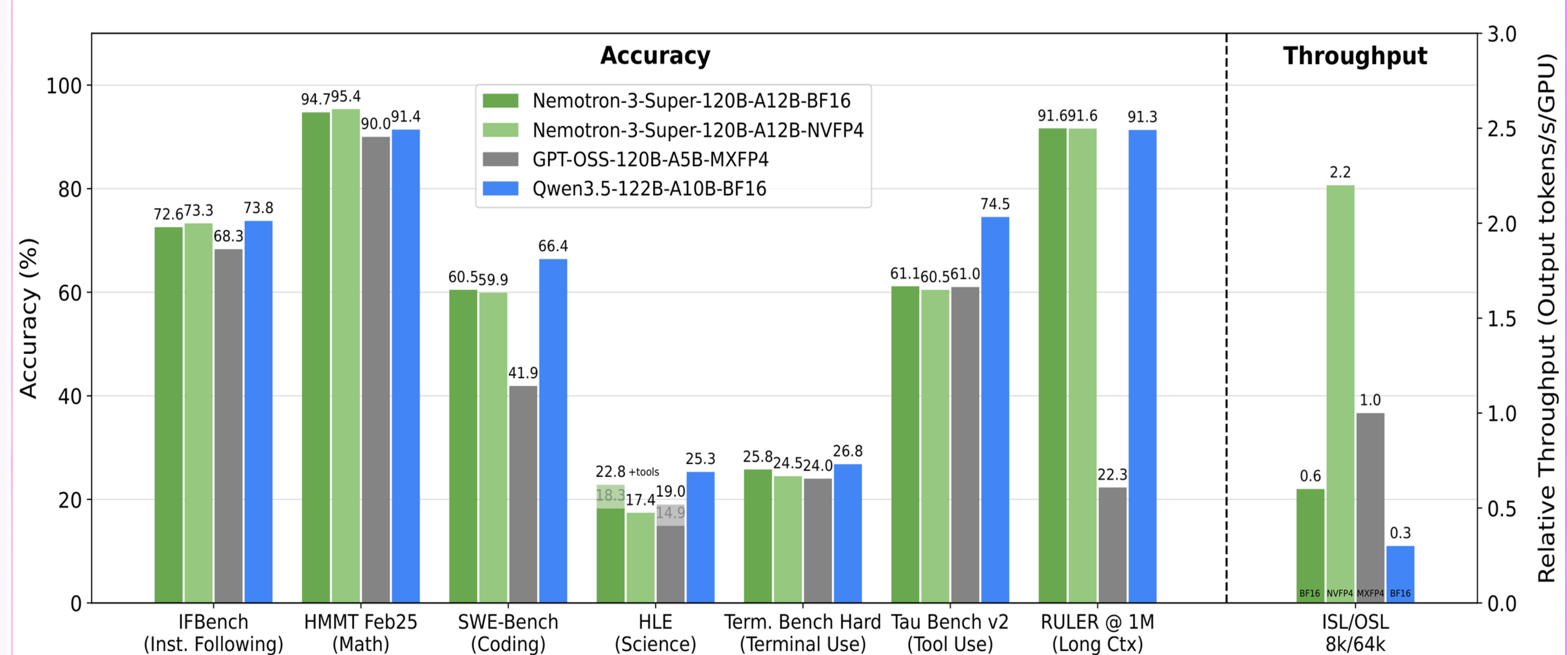


Figure 2 | Nemotron 3 Super layer pattern. Similar to Nemotron 3 Nano, we use a

## 3. RESULTS

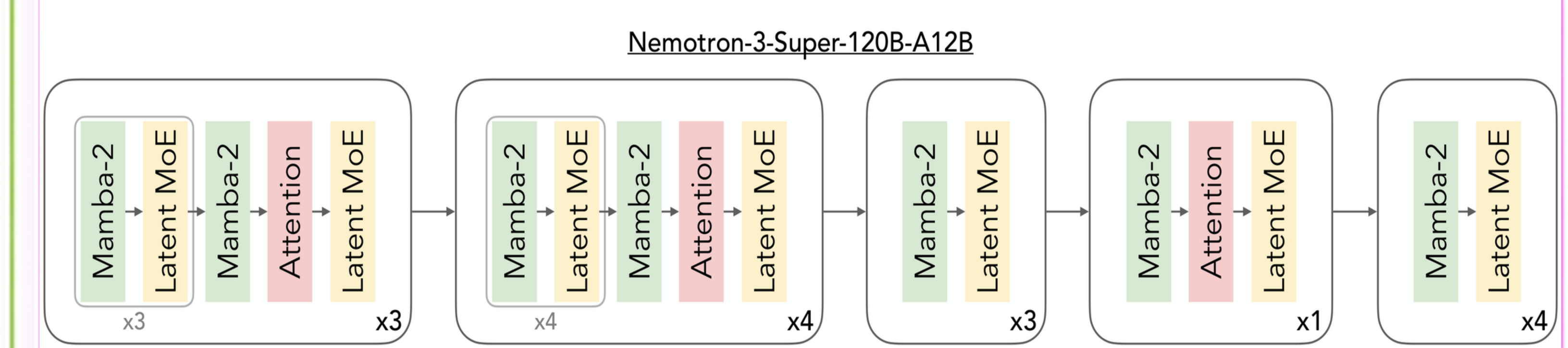


Figure 1 | Accuracy and throughput comparisons of Nemotron 3 Super with GPT-OSS-

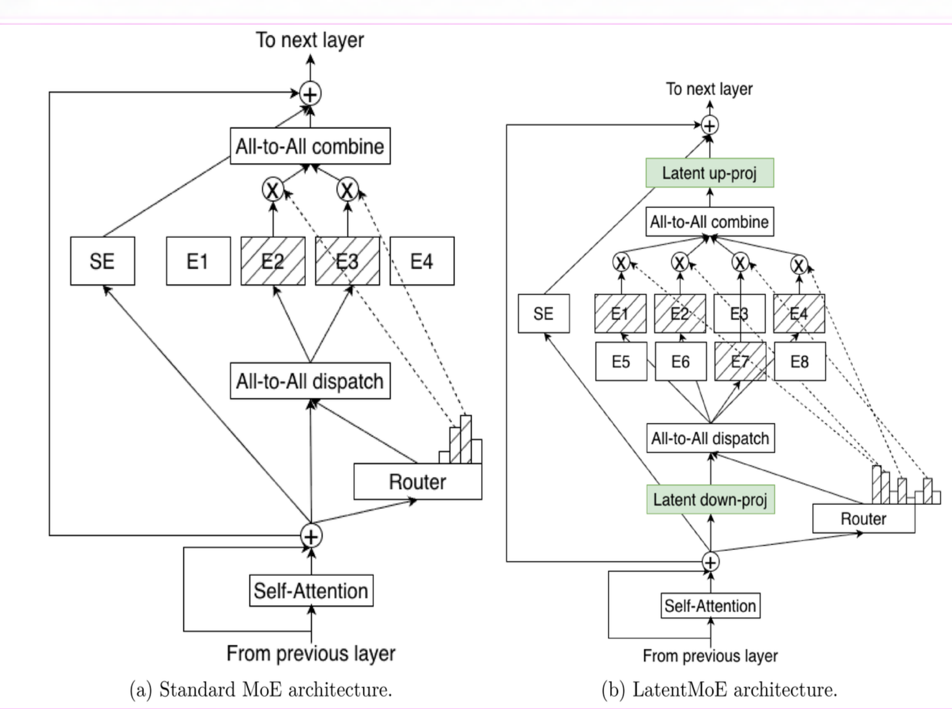


Figure 3 | Standard MoE vs. LatentMoE. In LatentMoE, tokens are projected from t

### BENCHMARK RESULTS (ACCURACY)

| | |
|---|---|
| HMMT Feb25 (Math) **94.7%** | SWE-Bench (Coding) **60.5%** |